



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 11

UNIT TEST - 2, AUGUST - 2024

COMPUTER SCIENCE

Time Allowed : 1.30 Hours]

[Max. Marks : 35

PART - I

1. Choose the correct answer.

10X1=10

- Omitting details inessential to the task and representing only the essential features of the task is known as
(a) specification (b) abstraction (c) composition (d) decomposition
- If $x = 6$ before the assignment $x := x+1$ after the assignment, the value of x is
(a) 5 (b) 4 (c) 3 (d) 7
- Suppose $u, v = 10, 20$ before the assignment. What are the values of u and v after the sequence of assignments?
1 $v := u$
2 $u := v$
(a) $u, v = 20, 20$ (b) $u, v = 20, 10$ (c) $u, v = 10, 20$ (d) $u, v = 10, 10$
- If C is false just before the loop, the control flows through
1 $S1$
2 while C
3 $S2$
4 $S3$
(a) $S1$ (b) $S1, S3$ (c) $S1, S2, S3$ (d) $S2, S3$
- How many times the loop is iterated?
 $x := 1$
while $x \leq 5$
 $x := x + 1$
(a) 4 (b) 5 (c) 6 (d) 0
- A loop invariant need not be true
(a) at the start of the loop (b) at the start of each iteration
(c) at the end of each iteration (d) at the start of the algorithm
- If $m \times a + n \times b$ is an invariant for the assignment $a, b := a + 8, b + 7$, the values of m and n are
(a) $m = 8, n = 7$ (b) $m = 7, n = -8$ (c) $m = 7, n = 8$ (d) $m = 8, n = -7$
- The division of main algorithm into functions is known as
(a) specification (b) abstraction (c) decomposition (d) composition
- Using this recursive definition

$$a^n = \begin{cases} 1 & \text{if } n = 0 \\ a \times a^{n-1} & \text{otherwise} \end{cases}$$

How many multiplications are needed to calculate a^{10} ?

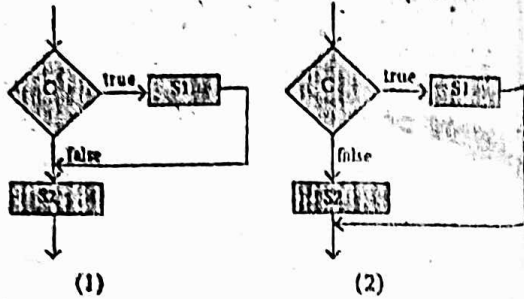
(b) 10

(c) 9

(d) 8

CH/II/C.S/1

10. If c is true, S1 is executed in both flow charts, but S2 is executed in



- a) (1) only b) (2) only
c) both (1) and (2) d) neither (1) or (2)

Part - II

II. Answer any 3 of the following:

3×2=6

11. Define an algorithm.
12. Specify a function to find the minimum of two numbers.
13. What is the difference between an algorithm and a program?
14. What is an invariant?

Part - III

III. Answer any THREE of the following:

3×3=9

15. What is abstraction?
16. What is the form and meaning of assignment statement?
17. What is case analysis?
18. Define a function to double a number in two different ways: (1) $n + n$, (2) $2 \times n$

Part - IV

IV. Answer all questions:

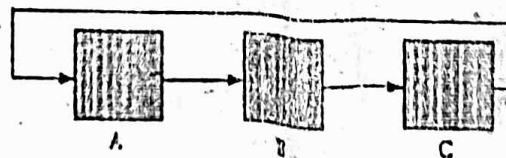
2×5=10

19. (a) Write the specification of an algorithm hypotenuse whose inputs are the lengths of the two shorter sides of a right-angled triangle, and the output is the length of the third side.

(OR)

- (b) Given two glasses marked A and B. Glass A is full of apple drink and glass B is full of grape drink. Write the specification for exchanging the contents of glasses A and B, and write a sequence of assignments to satisfy the Specification.?

20. (a) Write the specification and construct an algorithm to circulate the contents of the variables A, B and C as shown below: The arrows indicate that B gets the value of A, C gets the value of B and A gets the value of C..



(OR)

- (b) (i) Define a loop invariant.
- (ii) Does testing the loop condition affect the loop invariant? Why?



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